

# Result Tables

Table 1: Top-1 accuracy on PACS and VLCS, ResNet-18.

| Method       | Art          | Cartoon      | Photo        | Sketch       | Avg          | Caltech      | Labelme      | Sun          | Voc          | Avg          |
|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| ERM          | 76.53        | 74.80        | 94.95        | 67.89        | 78.54        | 96.87        | 65.03        | 71.62        | 71.60        | 76.28        |
| RSC          | 79.26        | 76.12        | 95.07        | 70.88        | 80.34        | 97.68        | 66.33        | 73.53        | 72.75        | 77.57        |
| CORAL        | 78.55        | 72.74        | 92.24        | 73.93        | 79.36        | 95.16        | 60.73        | 70.66        | 65.00        | 72.89        |
| MIXUP        | 79.79        | 72.81        | 93.11        | 63.97        | 77.42        | 93.44        | 60.25        | <b>74.36</b> | 69.87        | 74.48        |
| MMD          | 77.75        | 73.35        | 92.34        | 69.74        | 78.29        | 97.28        | 60.62        | 70.48        | 65.21        | 73.40        |
| SagNet       | 78.71        | 74.90        | 94.79        | 70.48        | 79.72        | 95.76        | 61.22        | 69.61        | 71.90        | 74.62        |
| SelfReg      | 79.46        | 76.32        | <b>95.93</b> | 71.60        | 80.83        | 92.53        | 62.56        | 70.27        | 72.15        | 74.38        |
| ARM          | 79.77        | 75.33        | 94.43        | <u>74.76</u> | 81.07        | 95.86        | 60.19        | 70.66        | 72.83        | 74.89        |
| EQRM         | 78.61        | 72.38        | 92.48        | 71.13        | 78.65        | 96.17        | 60.57        | 70.44        | 64.66        | 72.96        |
| SAGM         | 78.22        | 75.14        | 94.79        | 70.46        | 79.65        | 96.37        | 66.22        | 71.14        | 71.35        | 76.27        |
| <b>M2</b>    | <u>80.64</u> | <u>76.91</u> | 95.50        | 74.20        | <u>81.81</u> | <u>98.24</u> | <u>67.02</u> | 73.80        | <u>72.92</u> | <u>78.00</u> |
| <b>M2-CL</b> | <b>81.04</b> | <b>77.48</b> | <u>95.70</u> | <b>75.81</b> | <b>82.51</b> | <b>98.69</b> | <b>67.48</b> | <u>74.05</u> | <b>73.76</b> | <b>78.50</b> |

Table 2: Top-1 accuracy on PACS and VLCS, ResNet-50.

| Method       | Art          | Cartoon      | Photo        | Sketch       | Avg          | Caltech      | Labelme      | Sun          | Voc          | Avg          |
|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| ERM          | 83.54        | 78.47        | 96.51        | 73.64        | 83.04        | 97.01        | 62.22        | 71.79        | 76.09        | 76.78        |
| RSC          | 82.08        | <b>80.26</b> | 96.51        | 75.45        | 83.57        | <b>97.81</b> | 64.44        | 74.60        | 75.53        | 78.10        |
| CORAL        | 83.94        | 79.27        | 94.37        | <b>81.98</b> | 84.89        | 97.01        | 61.34        | 75.26        | 71.74        | 76.34        |
| MIXUP        | 85.16        | 77.40        | 96.23        | 78.26        | 84.26        | 96.94        | 62.33        | 73.40        | 72.32        | 76.25        |
| MMD          | 82.21        | 74.70        | 92.89        | 69.00        | 79.70        | 97.48        | 63.85        | 75.49        | 71.65        | 77.12        |
| SagNet       | 85.23        | 78.47        | 93.20        | 77.71        | 83.65        | 94.14        | 60.43        | 70.49        | 69.96        | 73.75        |
| SelfReg      | 83.33        | 79.06        | 93.16        | 77.24        | 83.20        | 92.49        | 60.73        | 66.80        | 73.58        | 73.40        |
| ARM          | 82.14        | 79.10        | 92.97        | 73.02        | 81.81        | 96.01        | 63.98        | 66.26        | 71.79        | 74.51        |
| EQRM         | 84.98        | 77.74        | 91.26        | 74.88        | 82.21        | 94.35        | 60.80        | 66.35        | 72.94        | 73.61        |
| SAGM         | 81.35        | 78.99        | 93.24        | 79.39        | 83.24        | 97.39        | 64.23        | 69.30        | 76.17        | 76.77        |
| <b>M2</b>    | <u>86.02</u> | 79.90        | <u>97.03</u> | 81.50        | <u>86.11</u> | 97.52        | <u>64.77</u> | <u>76.02</u> | <u>76.72</u> | <u>78.76</u> |
| <b>M2-CL</b> | <b>86.48</b> | <u>80.05</u> | <b>97.99</b> | <u>81.70</u> | <b>86.56</b> | <u>97.60</u> | <b>64.86</b> | <b>76.65</b> | <b>77.26</b> | <b>79.09</b> |

Table 3: Top-1 accuracy on Office-Home, ResNet-18.

| Method       | Art          | Clipart      | Product      | Real-World   | Avg          |
|--------------|--------------|--------------|--------------|--------------|--------------|
| ERM          | 56.12        | 47.88        | <u>72.47</u> | 73.58        | 62.51        |
| RSC          | 56.00        | 46.09        | <u>70.85</u> | 72.32        | 61.31        |
| CORAL        | 55.46        | 49.39        | 71.75        | 73.28        | 62.47        |
| MIXUP        | 56.37        | <u>49.81</u> | 70.87        | 73.61        | 62.66        |
| MMD          | 55.67        | <u>48.52</u> | 71.68        | 73.19        | 62.27        |
| SagNet       | 55.75        | 47.74        | 71.71        | 73.17        | 62.09        |
| SelfReg      | 57.56        | 49.69        | 72.43        | 73.72        | 63.35        |
| ARM          | 54.47        | 48.45        | 71.14        | <u>73.79</u> | 61.96        |
| EQRM         | 56.08        | 48.34        | 71.57        | 72.94        | 62.23        |
| SAGM         | 55.79        | 46.51        | 71.16        | 73.15        | 61.65        |
| <b>M2</b>    | <u>58.04</u> | 49.30        | <b>73.04</b> | 73.50        | <u>63.47</u> |
| <b>M2-CL</b> | <b>58.77</b> | <b>50.86</b> | 72.20        | <b>74.91</b> | <b>64.18</b> |

Table 4: Top-1 accuracy on Office-Home, ResNet-50.

| Method       | Art          | Clipart      | Product      | Real-World   | Avg          |
|--------------|--------------|--------------|--------------|--------------|--------------|
| ERM          | 63.43        | 51.32        | 76.45        | <u>78.13</u> | 67.33        |
| RSC          | 62.64        | 51.55        | 75.11        | <u>77.34</u> | 66.66        |
| CORAL        | 61.27        | <u>54.34</u> | 73.35        | 77.83        | 66.70        |
| MIXUP        | 63.23        | 51.91        | <u>77.10</u> | 76.39        | 67.16        |
| MMD          | 59.63        | 49.37        | <u>75.28</u> | 74.71        | 64.75        |
| SagNet       | 61.41        | 51.85        | 72.85        | 73.79        | 64.97        |
| SelfReg      | 59.95        | 51.22        | 71.28        | 74.52        | 64.24        |
| ARM          | 54.23        | 52.14        | 71.73        | 71.26        | 62.34        |
| EQRN         | 63.52        | 52.32        | 73.28        | 76.45        | 66.39        |
| SAGM         | 61.01        | 53.27        | 74.81        | 77.00        | 66.52        |
| <b>M2</b>    | <u>65.35</u> | 53.90        | 76.80        | <b>78.74</b> | <u>68.70</u> |
| <b>M2-CL</b> | <b>66.46</b> | <b>54.82</b> | <b>78.24</b> | 77.85        | <b>69.34</b> |

Table 5: Ablation study on PACS and VLCS, ResNet-18.

| pipe | r | drop | loss | A     | C     | P     | S     | PACS Avg | C     | L     | S     | V     | VLCS Avg |
|------|---|------|------|-------|-------|-------|-------|----------|-------|-------|-------|-------|----------|
| c    | 2 | -    | -    | 72.34 | 72.29 | 94.19 | 69.51 | 78.23    | 93.87 | 60.91 | 65.89 | 61.91 | 71.22    |
| c    | 4 | -    | -    | 74.38 | 72.74 | 90.13 | 69.71 | 77.21    | 94.14 | 59.01 | 67.82 | 65.42 | 69.62    |
| c    | 6 | -    | -    | 75.72 | 70.12 | 94.26 | 73.33 | 77.92    | 91.93 | 56.81 | 67.28 | 64.10 | 68.95    |
| c    | 2 | yes  | -    | 75.67 | 73.29 | 92.16 | 75.95 | 77.93    | 93.73 | 57.72 | 66.34 | 62.50 | 71.74    |
| c    | 4 | yes  | -    | 74.36 | 74.05 | 92.57 | 76.45 | 79.37    | 96.07 | 56.28 | 67.07 | 64.30 | 65.81    |
| c    | 6 | yes  | -    | 75.97 | 72.77 | 93.01 | 74.73 | 76.81    | 92.94 | 59.73 | 66.15 | 64.51 | 70.09    |
| p    | 2 | -    | -    | 76.09 | 55.88 | 68.35 | 60.92 | 64.40    | 95.60 | 56.97 | 66.25 | 73.15 | 73.96    |
| p    | 4 | -    | -    | 74.72 | 74.62 | 92.01 | 70.32 | 77.64    | 95.20 | 58.10 | 66.20 | 64.90 | 70.29    |
| p    | 6 | -    | -    | 75.38 | 69.99 | 91.21 | 69.47 | 77.77    | 95.23 | 56.81 | 69.13 | 65.34 | 70.55    |
| p    | 2 | yes  | -    | 74.61 | 72.39 | 87.08 | 73.95 | 78.88    | 95.73 | 58.33 | 68.44 | 63.74 | 70.49    |
| p    | 6 | yes  | -    | 76.73 | 73.98 | 93.41 | 73.56 | 79.84    | 93.19 | 56.05 | 66.62 | 64.24 | 70.43    |
| p    | 4 | yes  | -    | 76.39 | 75.98 | 95.58 | 74.49 | 79.15    | 94.54 | 59.77 | 67.26 | 74.55 | 73.14    |
| p    | 4 | yes  | yes  | 80.43 | 75.43 | 96.87 | 74.42 | 80.10    | 96.09 | 61.15 | 69.05 | 75.70 | 74.15    |

Table 6: Sensitivity analysis for  $\tau$  in M2-CL, ResNet-18.

| $\tau$ | PACS  | VLCS  |
|--------|-------|-------|
| 0.01   | 78.10 | 76.15 |
| 0.1    | 80.03 | 77.27 |
| 0.2    | 81.37 | 74.71 |
| 0.4    | 82.07 | 72.03 |
| 0.6    | 83.70 | 74.15 |
| 0.8    | 84.67 | 75.21 |
| 1.0    | 80.07 | 76.24 |
| 1.2    | 81.34 | 75.11 |
| 1.4    | 83.14 | 75.69 |
| 1.6    | 82.17 | 76.84 |
| 1.8    | 80.51 | 74.66 |
| 2.0    | 78.72 | 74.99 |
| 10.0   | 79.92 | 74.04 |
| 100.0  | 81.84 | 73.74 |

Table 7: Sensitivity analysis for  $\alpha$  in M2-CL, ResNet-18.

| $\alpha$ | PACS  | VLCS  |
|----------|-------|-------|
| 0.00     | 82.76 | 72.36 |
| 10e-5    | 80.41 | 73.56 |
| 10e-4    | 81.17 | 73.71 |
| 10e-3    | 82.29 | 74.38 |
| 10e-2    | 79.70 | 73.39 |
| 10e-1    | 64.04 | 55.96 |